

STOCK MARKET WIZARD

Analyzing the Philippine Stock Market through Machine Learning Models

November 2024

INTRODUCTION

The stock market, also referred to as the equity market or share bazaar, is a marketplace where shares of publicly held companies are bought and sold. In the Philippines, the [Philippine Stock Exchange \(PSE\)](#) is the central hub for stock trading. Understanding how the Philippine stock market operates enables us to make informed decisions about investments and economic trends. It is important to keep track of the stock market's value, as it has become a key platform for businesses to raise capital and serves as an indicator of a country's financial health.

One way to understand to predict stock market prices is through using machine learning algorithms. Stock market trend prediction is regarded as a crucial undertaking that requires careful consideration because, by making the right choices, accurate stock price predictions can result in substantial gains.

RESEARCH PROBLEM

Most investment strategies for Philippine stocks continue to rely heavily on fundamental analysis, technical indicators, or traditional econometric models.

However, there is a notable gap in the application of advanced machine learning algorithms specifically within the Philippine stock market. There is limited research regarding the performance of machine learning models in the Philippine stock market relative to larger, more developed markets.

Review of Related Literature

Machine learning algorithms outperformed traditional methods. They also concluded that SVM and ANN have the highest rank score results for the different algorithm classes.

L. Ryll and S. Seidens, "Evaluating the Performance of Machine Learning Algorithms in Financial Market Forecasting: A Comprehensive Survey," arxiv.org, Jun. 2019, Available: <https://arxiv.org/abs/1906.07786>

Review of Related Literature

Employed the following machine learning algorithms: Decision Tree, Random Forest, Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Support Vector Classifier (SVC), Naïve Bayes, K-Nearest Neighbors (KNN), Logistic Regression, and Artificial Neural Network (ANN); along with two deep learning algorithms: Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM)

Their results showed that each algorithm performed well when continuous data was used, though performance improved when binary data was applied, with accuracy increasing from 67% to 83%. It was also revealed that RNN and LSTM outperformed the machine learning models

M. Nabipour, P. Nayyeri, H. Jabani, S. S., and A. Mosavi, "Predicting Stock Market Trends Using Machine Learning and Deep Learning Algorithms Via Continuous and Binary Data; a Comparative Analysis," IEEE Access, vol. 8, pp. 150199–150212, 2020, doi: <https://doi.org/10.1109/access.2020.3015966>

Review of Related Literature

Kakarla et al. (2024) predicted stock prices using the Long Short-Term Memory (LSTM) algorithm, Convolutional Neural Network (CNN), and Artificial Neural Network (ANN). They applied these three algorithms to the Indian stock market, specifically MAANG and Divis Labs, and found that the neural network-based ANN yielded the best results, as it captures intricate, non-linear relationships and patterns. Their study revealed that while all three algorithms performed well on the dataset, the LSTM outperformed the others, with Root Mean Square Error (RMSE) and accuracy scores of 0.42 and 0.55, respectively.

K. P. Kakarla, C. Sai, and M. Kanchana, "Stock Price Prediction Using LSTM, CNN and ANN," Lecture notes in networks and systems, pp. 141–153, Jan. 2024, doi: https://doi.org/10.1007/978-981-99-8451-0_12.

Methodology

Datasets

The dataset utilized for this research consists of daily historical data from the [Philippine Stock Market \(PSE\)](#) and the [Philippine Stock Market Index \(PSEi\)](#). It contains 4 features:

- *Highest Price (High)*
- *Lowest Price (Low)*
- *Opening Price (Open)*
- *Volume*
- *Change*

These features reflect the trading activity of a selected stock or index within October 13, 2014 to October 11, 2024.

Methodology

Algorithm

1. Linear Regression
2. Support Vector Machine (SVM)
3. Long Short-Term Memory (LSTM)

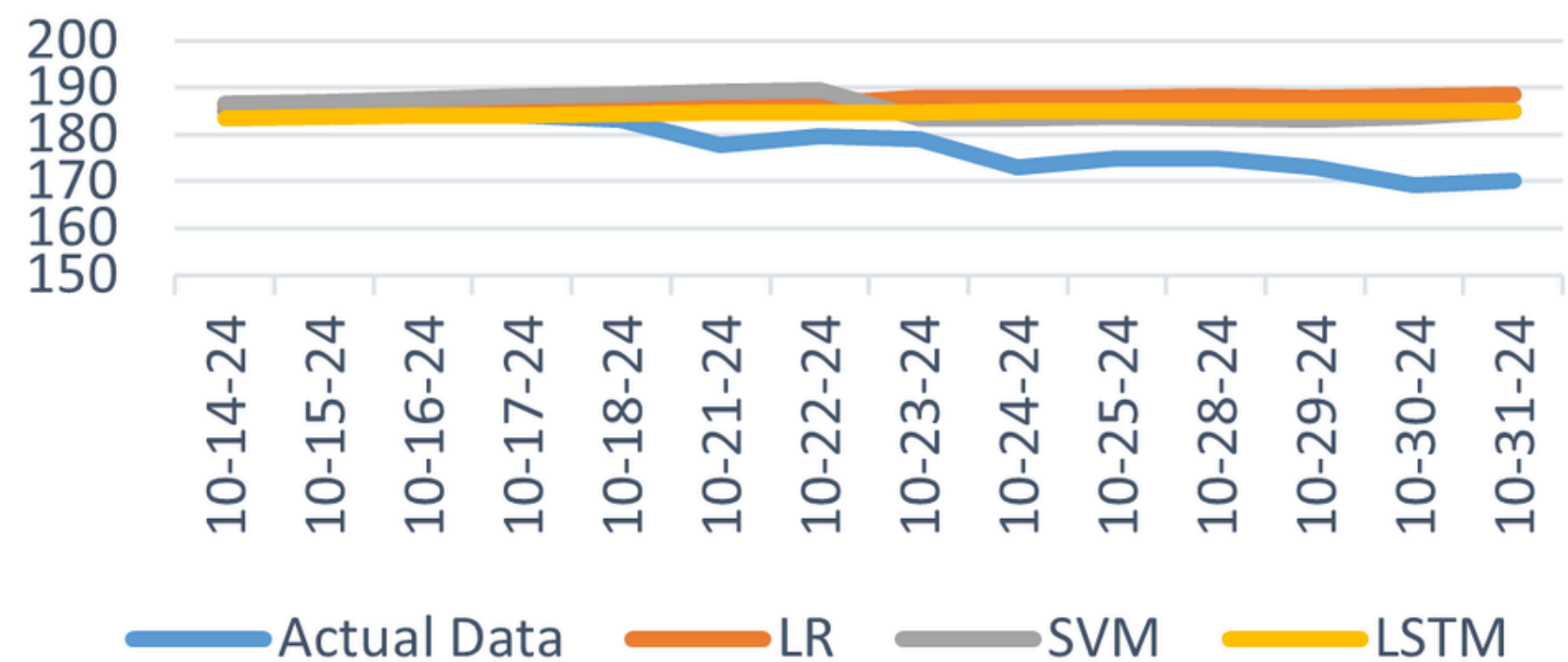
Results and Discussion

Philippine Stock Market			
	LR	SVM	LSTM
Mean Squared Error (MSE)	7.76	8.36	14.41
Mean Absolute Error (MAE)	2.05	2.22	2.98
Root Mean Squared Error (RMSE)	2.79	2.89	3.80
R-squared	0.9609	0.9575	0.9267

Philippine Stock Market Index			
	LR	SVM	LSTM
Mean Squared Error (MSE)	3956.55	3945.85	10981.80
Mean Absolute Error (MAE)	49.42	49.22	82.22
Root Mean Squared Error (RMSE)	62.90	62.82	104.79
R-squared	0.9526	0.9514	0.8647

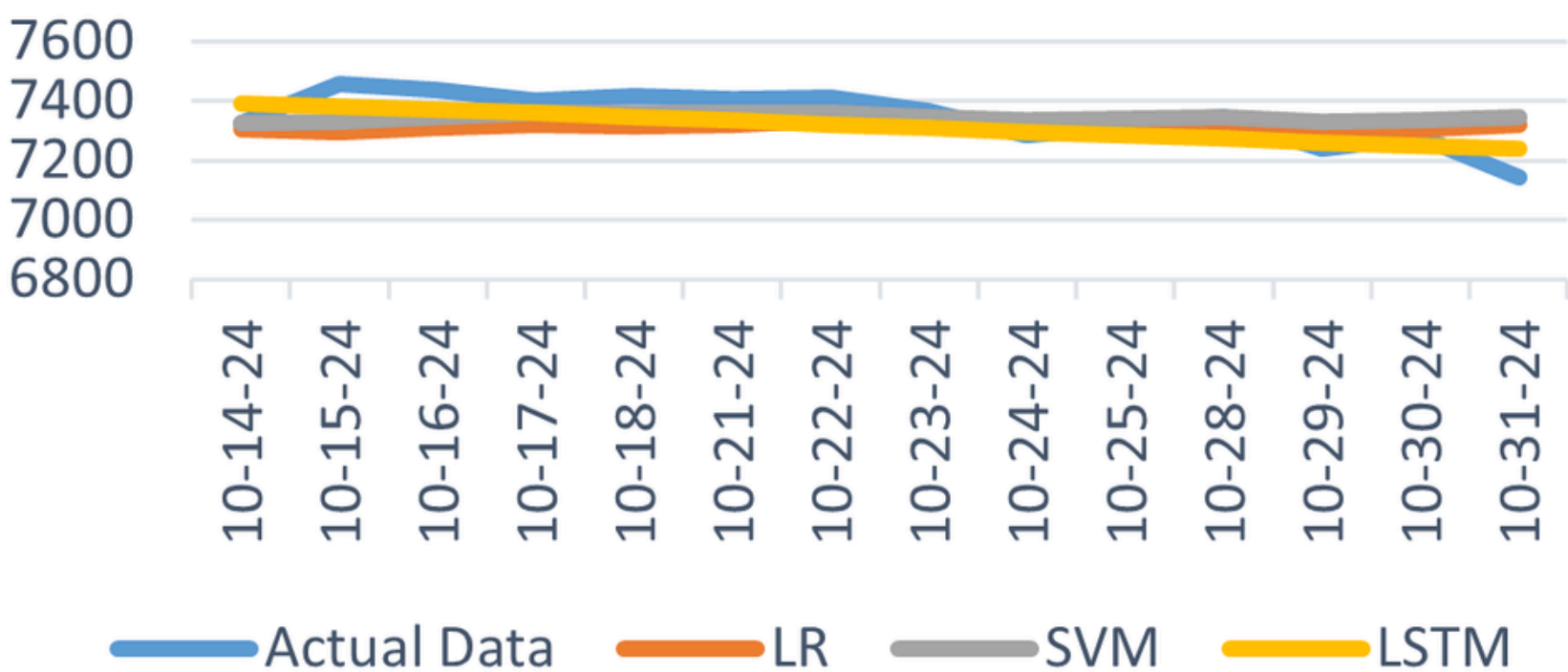
Results and Discussion

Actual vs. Predicted



PSE

Actual vs. Predicted



PSEi



Conclusions

Group 2

CCMACLRL

Thank you.

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